Shadow Crew — Technical Sheet

Tech

One self-contained HTML file (about 27 KB), no libraries, images or network. HTML5 Canvas 2D rendering at up to 2x device pixel ratio, mobile-first touch input, Web Audio API synthesized music and effects (optional, starts on first tap), localStorage for record, difficulty and mute.

Systems

Station: 8 rooms around a central hallway, door-based collision and waypoint pathing for bots.

Tasks: each crewmate gets 4 of 7 consoles; bots do theirs in 4 to 6 seconds, you hold USE for 3.

Impostor AI: hunts a crewmate, only kills when no crew can see it, with a kill cooldown and a station-wide gap between kills. A nearby crewmate may witness a kill and call a meeting at once.

Evidence: every second, each crewmate logs who it can see and where. At a meeting, crew bots report sightings near the body in the 20 seconds before death plus alibis; impostors add lies and cover.

Voting: each bot builds suspicion from statements, distrusts anyone who accused it falsely, and only votes when confident; plurality over skip ejects.

Rendering: neon rooms with glow, parallax starfield, animated crew, vision fog, particles and screenshake.

Audit 1: Balance (simulated play)

A bot played your seat in 150 full games per difficulty. First version: crew won 19% on Easy and 0 to 1% on Normal and Hard, because impostors killed every ~25 s and bodies went unfound. Fixes: longer kill cooldowns (50/70/55 s), 25 s gap between any two kills, bots spot bodies from further away, 4 tasks per crewmate with quicker bot tasks, and crew bots need stronger evidence before voting. After the fixes, the passive bot player wins 65% on Easy, 29% on Normal and 26% on Hard; games last about 2.5 minutes. A human who sticks with the crew and reads the clues does better than the bot.

Audit 2: Bugs

A headless DOM harness played full games with meetings and votes. Fixed: the end screen could fire twice when you were voted out during a meeting; meeting overlay now locks votes after one tap; the task list no longer rebuilds every frame (it caused scroll jitter on phones); audio failure no longer blocks play.

Audit 3: Player experience

Added "You noticed" lines so your own sightings show up at meetings, a clear message when nobody saw anything, a toast with how many impostors remain after each vote, the vote tally after each ejection, yellow highlighting on your own task rooms and consoles, and a list of the impostors on the end screen.